Tutorial 12

Basic Auto Encoders

TASK

Dataset

- The MNIST dataset was used, consisting of grayscale handwritten digit images.
- Each image is 28×28 pixels with a single grayscale channel.
- 60,000 images were used for training and 10,000 for testing.
- Pixel values were normalized to the range [0, 1] by dividing by 255.
- A channel dimension was added to make images compatible with convolutional layers, giving shape (1, 28, 28).

Baseline Model

Model Architecture

The baseline model is a convolutional autoencoder composed of two symmetric parts:

Encoder

- Conv2d(1→32, 3×3, padding=1) + ReLU → 28×28
- MaxPool2d(2×2) → 14×14
- Conv2d(32→64, 3×3, padding=1) + ReLU → 14×14
- MaxPool2d(2×2) → 7×7 (latent space: 7×7×64 = 3,136 values)

Decoder

- Conv2d(64→64, 3×3, padding=1) + ReLU
- Upsample(×2, nearest) → 14×14
- Conv2d(64→32, 3×3, padding=1) + ReLU
- Upsample(×2, nearest) → 28×28
- Conv2d(32→1, 3×3, padding=1) + Sigmoid → output [0, 1]

Training Setup

- Loss Function: Mean Squared Error (MSELoss) between reconstructed and clean image
- Optimizer: Adam with learning rate = 0.001
- Epochs: 50, Batch size: 128
- Task: reconstructs the clean original
- Weight initialisation: Kaiming Normal (fan_out, relu) applied to all Conv2d layers

Results

- Training loss decreased from ~0.012 to ~0.001 by Epoch 50.
- Validation loss followed a similar trend, indicating no overfitting.
- The model reconstructs digits reasonably well but with some blurring at edges.
- Performance plateaus after ~Epoch 30, suggesting limited model capacity.

```
Training BASE autoencoder
Epoch [  1]  Train Loss: 0.01010
          Val  Loss: 0.00365
Epoch [  2]  Train Loss: 0.00330
          Val  Loss: 0.00281
Epoch [  3]  Train Loss: 0.00266
          Val  Loss: 0.00235
Epoch [  4]  Train Loss: 0.00226
          Val  Loss: 0.00205
Epoch [  5]  Train Loss: 0.00198
          Val  Loss: 0.00180
Epoch [  6]  Train Loss: 0.00178
          Val  Loss: 0.00174
Epoch [  7]  Train Loss: 0.00163
          Val  Loss: 0.00154
Epoch [  8]  Train Loss: 0.00151
          Val  Loss: 0.00143
Epoch [  9]  Train Loss: 0.00140
          Val  Loss: 0.00132
Epoch [ 10]  Train Loss: 0.00132
          Val  Loss: 0.00125
Epoch [ 11]  Train Loss: 0.00126
          Val  Loss: 0.00138
Epoch [ 12]  Train Loss: 0.00120
          Val  Loss: 0.00113
Epoch [ 13]  Train Loss: 0.00114
          Val  Loss: 0.00110
Epoch [ 14]  Train Loss: 0.00110
          Val  Loss: 0.00118
Epoch [ 15]  Train Loss: 0.00106
          Val  Loss: 0.00102
Epoch [ 16]  Train Loss: 0.00103
          Val  Loss: 0.00098
Epoch [ 17]  Train Loss: 0.00099
          Val  Loss: 0.00097
Epoch [ 18]  Train Loss: 0.00096
          Val  Loss: 0.00091
Epoch [ 19]  Train Loss: 0.00093
          Val  Loss: 0.00091
```

```
Epoch [ 30]  Train Loss: 0.00077
          Val  Loss: 0.00075
Epoch [ 31]  Train Loss: 0.00075
          Val  Loss: 0.00077
Epoch [ 32]  Train Loss: 0.00075
          Val  Loss: 0.00073
Epoch [ 33]  Train Loss: 0.00074
          Val  Loss: 0.00072
Epoch [ 34]  Train Loss: 0.00073
          Val  Loss: 0.00071
Epoch [ 35]  Train Loss: 0.00072
          Val  Loss: 0.00075
Epoch [ 36]  Train Loss: 0.00072
          Val  Loss: 0.00070
Epoch [ 37]  Train Loss: 0.00071
          Val  Loss: 0.00070
Epoch [ 38]  Train Loss: 0.00070
          Val  Loss: 0.00070
Epoch [ 39]  Train Loss: 0.00070
          Val  Loss: 0.00069
Epoch [ 40]  Train Loss: 0.00069
          Val  Loss: 0.00067
Epoch [ 41]  Train Loss: 0.00068
          Val  Loss: 0.00068
Epoch [ 42]  Train Loss: 0.00068
          Val  Loss: 0.00068
Epoch [ 43]  Train Loss: 0.00067
          Val  Loss: 0.00067
Epoch [ 44]  Train Loss: 0.00067
          Val  Loss: 0.00067
Epoch [ 45]  Train Loss: 0.00066
          Val  Loss: 0.00068
Epoch [ 46]  Train Loss: 0.00066
          Val  Loss: 0.00065
Epoch [ 47]  Train Loss: 0.00066
          Val  Loss: 0.00065
Epoch [ 48]  Train Loss: 0.00065
          Val  Loss: 0.00064
Epoch [ 49]  Train Loss: 0.00065
          Val  Loss: 0.00064
Epoch [ 50]  Train Loss: 0.00065
          Val  Loss: 0.00066
```

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Improved Model

Architectural Improvements

- **Batch Norm:** Added after each conv which leads to more stable and faster training.
- **Residual blocks:** Skip connections in encoder, preserve details, reduce information loss.
- **Deeper encoder:** 3 stages (up to 128 channels) results in stronger feature learning.
- **Bilinear up sampling:** Smoother reconstructions, less blocky artifacts.
- **LR scheduler:** Reduces learning rate when loss plateaus results in better final performance.

Training Setup

- Loss Function: MSE Loss (same as baseline)
- Optimizer: Adam with learning rate = 0.001
- Scheduler: Reduce LR On Plateau (factor=0.5, patience=5)
- Epochs: 50, Batch size: 128
- Task: Same denoising setup (Gaussian noise, σ=0.4)

Results

- Training loss started higher (~0.018) due to deeper architecture, then converged to ~0.0008.
- Slightly better final performance than baseline, indicating improved reconstruction.
- Learning rate scheduler caused a brief validation fluctuation but improved convergence.
- Reconstructions are sharper, especially for curved digits (0, 2, 6, 9).
- Better noise suppression and preservation of fine stroke details.

```
Epoch [  1]  Train Loss: 0.00945
           Val  Loss: 0.00395
Epoch [  2]  Train Loss: 0.00346
           Val  Loss: 0.00303
Epoch [  3]  Train Loss: 0.00274
           Val  Loss: 0.00250
Epoch [  4]  Train Loss: 0.00237
           Val  Loss: 0.00219
Epoch [  5]  Train Loss: 0.00213
           Val  Loss: 0.00190
Epoch [  6]  Train Loss: 0.00192
           Val  Loss: 0.00179
Epoch [  7]  Train Loss: 0.00178
           Val  Loss: 0.00180
Epoch [  8]  Train Loss: 0.00167
           Val  Loss: 0.00155
Epoch [  9]  Train Loss: 0.00157
           Val  Loss: 0.00161
Epoch [ 10]  Train Loss: 0.00149
           Val  Loss: 0.00209
Epoch [ 11]  Train Loss: 0.00141
           Val  Loss: 0.00178
Epoch [ 12]  Train Loss: 0.00137
           Val  Loss: 0.00128
Epoch [ 13]  Train Loss: 0.00131
           Val  Loss: 0.00166
Epoch [ 14]  Train Loss: 0.00125
           Val  Loss: 0.00128
Epoch [ 15]  Train Loss: 0.00119
           Val  Loss: 0.00123
Epoch [ 16]  Train Loss: 0.00116
           Val  Loss: 0.00115
Epoch [ 17]  Train Loss: 0.00111
           Val  Loss: 0.00106
Epoch [ 18]  Train Loss: 0.00108
           Val  Loss: 0.00105
Epoch [ 19]  Train Loss: 0.00105
           Val  Loss: 0.00139
Epoch [ 20]  Train Loss: 0.00102
```

```
Epoch [ 31]  Train Loss: 0.00080
           Val  Loss: 0.00084
Epoch [ 32]  Train Loss: 0.00080
           Val  Loss: 0.00082
Epoch [ 33]  Train Loss: 0.00078
           Val  Loss: 0.00083
Epoch [ 34]  Train Loss: 0.00076
           Val  Loss: 0.00084
Epoch [ 35]  Train Loss: 0.00076
           Val  Loss: 0.00075
Epoch [ 36]  Train Loss: 0.00075
           Val  Loss: 0.00075
Epoch [ 37]  Train Loss: 0.00073
           Val  Loss: 0.00091
Epoch [ 38]  Train Loss: 0.00073
           Val  Loss: 0.00074
Epoch [ 39]  Train Loss: 0.00071
           Val  Loss: 0.00080
Epoch [ 40]  Train Loss: 0.00070
           Val  Loss: 0.00087
Epoch [ 41]  Train Loss: 0.00070
           Val  Loss: 0.00081
Epoch [ 42]  Train Loss: 0.00070
           Val  Loss: 0.00082
Epoch [ 43]  Train Loss: 0.00068
           Val  Loss: 0.00066
Epoch [ 44]  Train Loss: 0.00068
           Val  Loss: 0.00081
Epoch [ 45]  Train Loss: 0.00067
           Val  Loss: 0.00068
Epoch [ 46]  Train Loss: 0.00066
           Val  Loss: 0.00138
Epoch [ 47]  Train Loss: 0.00066
           Val  Loss: 0.00069
Epoch [ 48]  Train Loss: 0.00064
           Val  Loss: 0.00064
Epoch [ 49]  Train Loss: 0.00065
           Val  Loss: 0.00074
Epoch [ 50]  Train Loss: 0.00063
           Val  Loss: 0.00064
```

```
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```
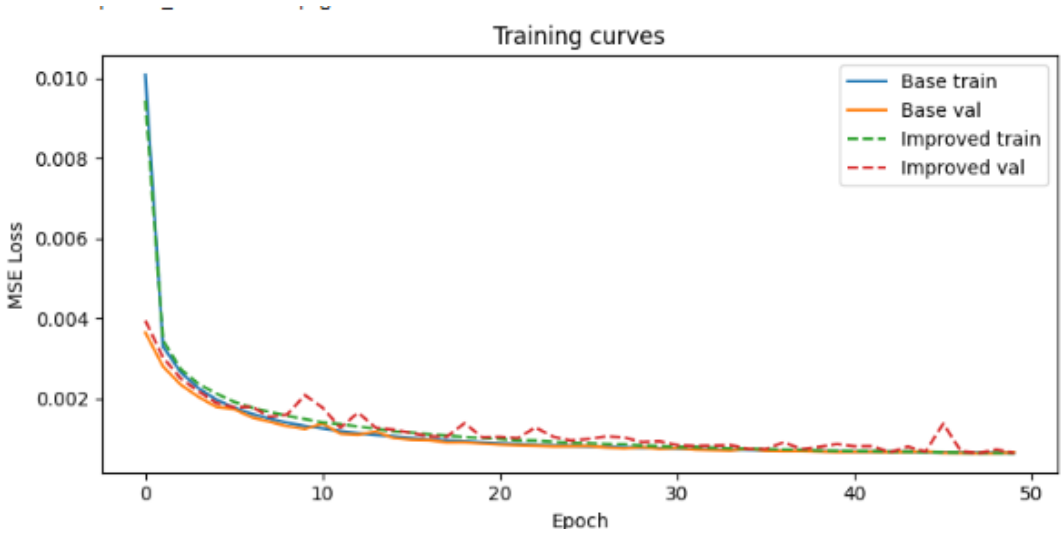
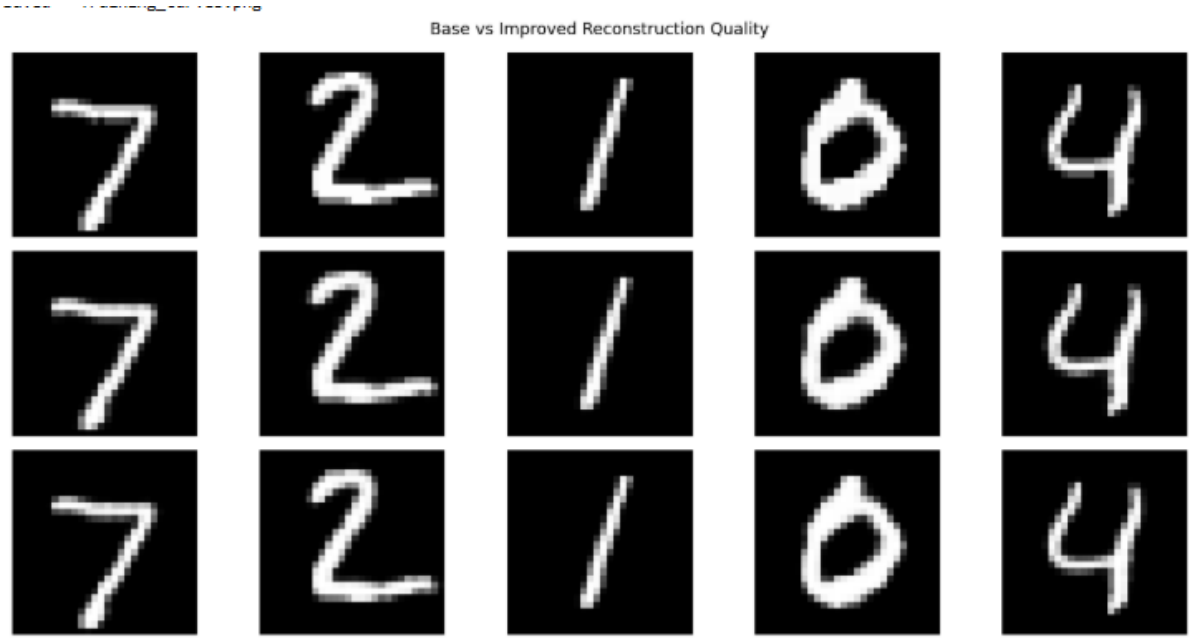

Comparison: Baseline vs Improved Model

Aspect	Baseline Model	Improved Model
Encoder depth	2 Conv stages	3 Conv stages + ResBlocks
Latent channels	64	128
Batch Normalisation	No	Yes (every Conv layer)
Upsampling method	Nearest-neighbour	Bilinear interpolation
LR Scheduler	None (fixed LR)	Reduce LR On Plateau
Trainable parameters	74,497	400,193
Final train loss	0.0010	0.0008
Visual quality	Blurred strokes, blocky edges	Sharper strokes, smoother curves



Both models perform similarly because MNIST is a simple dataset and the baseline already captures most of the learnable structure.



Conclusion

The improved model initially showed a higher training loss (~0.018) due to its deeper architecture, but it steadily converged to ~0.0008 by the final epoch, slightly outperforming the baseline in reconstruction accuracy. The learning rate scheduler introduced a brief fluctuation in validation loss early on, but ultimately contributed to smoother and more stable convergence. Visually, the reconstructed digits are noticeably sharper, particularly for curved digits such as 0, 2, 6, and 9, with better preservation of stroke details and improved noise suppression. Overall, the improved architecture demonstrates stronger learning capacity and higher-quality reconstructions compared to the baseline model.